

THE TRAINING DATA LESSON • LESSON 2 OF 6

What's in the Playlist?



How AI learns, and where bias comes from. An AI is a music app and its training data is the playlist. A one-vibe playlist gets one-vibe answers. Stands alone.

AT A GLANCE

TIME	MATERIALS	TECH NEEDED	PREP
45 min	Playlist Check worksheet, 2 pages, 1 per student — each fills their OWN sheet (typeable PDF copy included) · Slides (optional; script in the notes) · board	None — runs fully unplugged. Optional: a projector for the deck	Print worksheets. That's it.

OBJECTIVES

- Explain how an AI learns: **input** → **patterns** → **output** — prediction from data, not magic. (*Exit Ticket #1*)
- Name who a dataset serves well and who it leaves out; link that gap to real AI bias. (*Gap Report; Exit #2*)
- Discuss data gaps with a partner: say their idea back, then add to it. (*Repeat, Then Add*)

STANDARDS ALIGNMENT

CSTA 2-IC-21

ISTE 1.1.d

CCSS ELA SL.6.1

Full text: Teacher Guide, p.3

DIFFERENTIATION

- Support:** Gray sentence starters on the worksheet; landmark-card tip in the Teacher Guide, p.2.
- Extension:** Design a 12-card playlist that serves every request on the sheet, then defend the tradeoffs. Model answer: Teacher Guide, p.3.
- ELL:** Fit marks are check boxes, not sentences. New cards can be in any language, which fixes a real gap.
- No-device classrooms:** Built in — the lesson is paper, pairs, and one loud imaginary music app.

WARM-UP — THE ONE-PLAYLIST APP (7 MIN)

You are **AutoMix**, the school's new music app. Your playlist holds one kind of song: loud, fast party tracks. Take 3–4 requests and serve every one with total confidence:

"Something chill to study to?" → "Here's 'TURBO TURBO.' It is not chill. It's what I've got."

"A song for grandma's birthday dinner?" → "Got it. 'VOLUME WAR.' Grandma loves bass drops, probably."

"A slow song for the last dance?" → "'NEON STAMPEDE,' played at... the exact same speed."

Then ask: **"Is AutoMix bad at picking songs — or is something else broken?"** Fish for it. The app is doing its job. The *playlist* is the problem.

If nobody gets there: "Could AutoMix play a slow song if it had one?" Yes — the app works. So what's missing?

Bridge: "If you did the genie lesson, you saw an AI grant what you SAID. If you didn't, no problem. Here's today's question: where do an AI's answers come from in the first place? What an AI 'knows' comes from a playlist of examples. That's its training data. It can remix the playlist. It can't serve what was never in it. Today you'll BE the app."

Transition: "So what IS in an AI's playlist, and what does it do with it? Three words on the board."

DIRECT INSTRUCTION — NO MAGIC IN THE MACHINE (8 MIN)

INPUT — the playlist. Training data: every example the AI studied before you met it. Nobody types "knowledge" in.

PATTERNS — the training. It works through the playlist and gets good at ONE thing: guessing what usually goes with what.

OUTPUT — the mix. It predicts what best matches those patterns. Its best guess, not a fact it looked up. It can sound sure and still be off.

Say it plainly (board: INPUT → PATTERNS → OUTPUT): "No magic in the machine. An AI doesn't KNOW things the way you do. It predicts them from its playlist. So two rules run everything: a GAP in the playlist is a gap in the AI, and a LEAN in the playlist leans its answers. That lean has a name: bias."

Quick check: "A photo AI's playlist is almost all golden retrievers. A husky walks in — what's it going to call it?" (Golden retriever — its best guess. Gap in, gap out.)

If a student says "husky": "Has this AI ever seen one? It can only pick from the patterns it has."

Transition (hand out Playlist Check): "Pair up. Read the gray box together. It names Partner A and your jobs."

ACTIVITY — PLAYLIST CHECK (20 MIN)

- Setup (2 min):** Partner A is the partner whose first name comes first in the alphabet. A starts as the **App**, B as the **Caller**. The 12 cards ARE the app's training data. First, each student reads the cards and fills in the pattern line under them (30 s). Rules: serve only from the playlist, never say no, one honest Fit box each.
If there's an odd number: a trio. One App, two Callers taking turns; swap the App twice.
Transition: "Caller, read #1 out loud. App — you can't say no."
- Serve the Caller (8 min):** Six requests (Priya, Nia, Tomás, Keisha, Malik, Jordan). App names the card; each checks their own Fit box (they may differ). Swap at #4. #7 is a live request from the pair beside them.
If a pair can't agree on the Fit: each checks their own — disagreement is data. Save it for the debrief.
Transitions: (#3) "Swap jobs: new App, new Caller. On to #4." (#7) "Pair closer to the door, you go first: make up ONE request for the other pair's app. Then switch." (after #7) "Pens on page 2. Say who it let down."
- Playlist Gap Report (7 min):** Each student writes who the playlist serves well and who it leaves out. Then Repeat, Then Add: Partner B names a gap; A says it back ("I heard you say...") and adds ("Building on that..."). Switch. Each writes their partner's gap and their own add.
If a pair writes "it needs more songs": ask WHO asked for something that wasn't there.
Transition: "Three minutes. Fix the playlist — invent the cards that are missing."
- Fix the Playlist (3 min):** Invent new cards that fix the biggest gaps — at least 2. Push past "add one slow song": which requests still have nothing?
If every new card is a slow song: "Tomás still has nothing. What does he need?"
Transition: "Pens down. Eyes up. Who did your playlist let down? Hands."

DEBRIEF — THE REAL PLAYLISTS (5 MIN)

2–3 pairs name one gap (1 min). Three real playlists: voices, faces, feeds (3 min; slide 9 notes). Big question (1 min): **"Nobody put bad songs in your playlist. So where did all that bias come from?"**

If nobody lands it: what got collected, and what didn't. "On purpose?" Usually not; teams grab the easiest data.

Transition: "Back to your own sheet. The bottom of page 2 is yours alone — five minutes."

EXIT TICKET (5 MIN)

Worksheet p.2, solo: the two exit questions (slide 11 shows them word for word). Key: Teacher Guide p.2.

If a student writes "the AI learns": "Learns from what? Say where the patterns came from."